Semi-Automated Reverse Engineering of Simple Mechanical Parts into Editable Parametric CAD Models

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Abstract

In circular-design manufacturing, reclaiming and remanufacturing existing components requires reliably converting 3D scan data into editable CAD models. We present a semi-automated reverse engineering workflow for simple mechanical parts such as blocks, flanges, and shafts that extracts geometric primitives (planar faces and cylindrical surfaces) from laser scans using RANSAC fitting and normal-based region-growing. Our six-stage pipeline comprises: (1) high-fidelity scan acquisition; (2) outlier removal and ICP alignment; (3) sequential RANSAC plane and cylinder detection; (4) region-growing refinement; (5) geometric parameter computation; and (6) automated Siemens NX reconstruction via the NX Open API.

Applied to three representative parts, the method achieves dimensional errors less than 0.5 mm with over 70% of modeling steps fully automated. Remaining manual interventions, seed-point selection and occluded-face definition are analyzed, and enhancements (automatic symmetry detection, multi-view fusion) are proposed to realize full end-to-end automation. This proof-of-concept demonstrates that a classical, training-free feature-extraction approach can deliver editable CAD models that support circular-economy goals of component reuse and resource efficiency.

1.Introduction

Circular-design manufacturing seeks to close the materials loop by reclaiming end-of-life components, digitizing their geometry, and remanufacturing them with minimal waste [1]. A critical enabler of this vision is reverse engineering, transforming raw 3D scans into editable, parametric CAD models. In practice, however, engineers often spend 15–20 minutes per part manually tracing dense point clouds achieving tolerances no better than ±0.5 mm which limits throughput and introduces human error[2].

Recent machine-learning frameworks (e.g., Point2CAD) can segment scans automatically but depend on large annotated datasets and yield non-parametric meshes unsuitable for direct feature-based CAD editing [3]. By contrast, classical geometry techniques notably RANSAC plane- and cylinder-fitting combined with normal-based region-growing can robustly extract the geometric primitives (planar faces and cylindrical surfaces) that constitute most mechanical parts, without training data. These primitives map directly to CAD "features" (extrudes, revolves), enabling a semi-automated, fully parametric reconstruction in Siemens NX via the NX Open API [4]. This decision is supported by prior work such as Yang and Förstner [8], who demonstrated that classical methods like RANSAC, when combined with the Minimum Description Length (MDL) principle, can robustly extract planes from noisy laser scans without requiring annotated training data.

In this work, we position our semi-automated reverse-engineering workflow at the intersection of Geometric Feature Extraction, Parametric CAD Automation, and Circular-Design Manufacturing, demonstrating how classical feature-based methods can bridge these domains for efficient, high-fidelity part reuse.

1.1 Research Question:

To what extent can planar and cylindrical primitives extracted from 3D laser scans via RANSAC fitting and normal-based region-growing be reconstructed in Siemens NX as fully parametric CAD models with dimensional error under 0.5 mm, and which stages of this semi-automated pipeline still require manual intervention?

1.2 Areas of Relevance and Contribution diagram

Figure 1 positions the semi-automated reverse-engineering workflow at the nexus of three core research areas Geometric Feature Extraction, Parametric CAD Automation, and Circular-Design Manufacturing each represented by a white bubble. Surrounding these essentials are the yellow "useful" enhancers: Adaptive Parameter Tuning improves the robustness of primitive detection, Symmetry & Occlusion Handling fills in missing geometry for complete CAD reconstruction, and Multi-Scan Fusion ensures full part coverage by merging multiple scan views. Finally, the blue contribution rings draw our focus outward to the high-level impacts enabled by this work: Editable Parametric CAD Models ready for design modification, Rapid Component Reuse in new assemblies, and Resource & Waste Reduction in manufacturing. Together, these layers clarify not the sequential pipeline steps, but rather the intellectual territory we occupy, the techniques that bolster our approach, and the sustainability benefits we deliver.

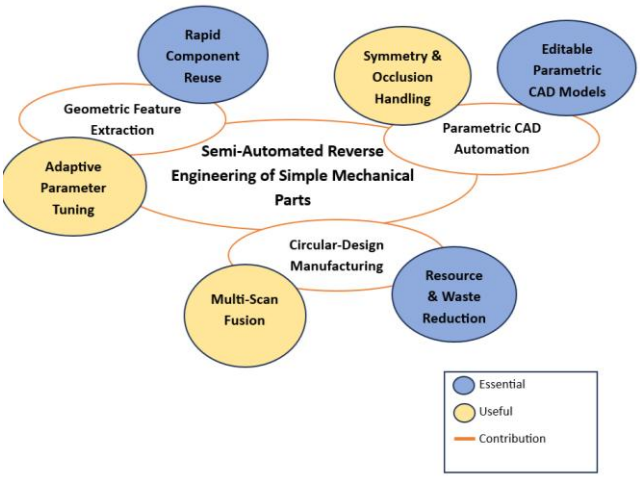
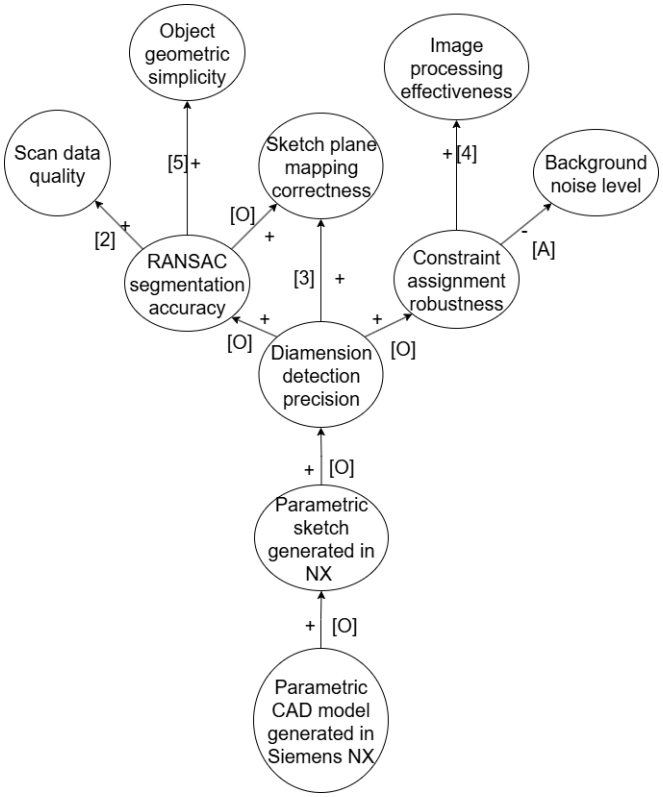


Figure 1: ARC diagram for feature-based reverse engineering in circular design

1.3 Reference Model



[O]: Own experimental findings
[A]: Logical assumptions
[2], [3], [4], [5]: References to relevant literature

Figure 2: Reference model for feature-based reverse engineering in circular design

The reference model in figure 2 illustrates the key factors influencing the automated generation of parametric CAD models from 3D scan data in Siemens NX. The model follows a bottom-up structure, starting from technical inputs and progressing toward the final outcome.

At the foundation is scan quality, which is affected by part complexity, scanning technology (e.g., Ametek HandySCAN), and environmental conditions such as lighting and background noise. These elements determine the clarity, resolution, and completeness of the scanned point cloud. Pre-processing quality builds upon scan quality and reflects operations like noise filtering, alignment, and mesh simplification. Tools such as Open3D and MeshLab are commonly used to improve point cloud structure before feature detection.

Next, feature extraction is performed using geometry fitting techniques such as RANSAC and region growing. The accuracy of this step depends on both pre-processing quality and segmentation reliability. A well-structured feature extraction process enables more accurate detection of dimensions.

Sketch plane mapping correctness and constraint assignment robustness are critical to generating clean, editable sketches in Siemens NX. These depend on both segmentation outcomes and the effectiveness of image processing. High-quality constraints (e.g., concentricity, horizontality) ensure that the sketch remains fully parametric and modifiable. At the upper levels, dimension detection precision directly affects the success of generating a valid parametric sketch, which then enables creation of a fully defined parametric CAD model. Each link in the model represents a causal relationship, where improvements in lower-level technical stages positively influence higher-level reconstruction outcomes.

Overall, this reference model defines the interdependencies that affect the quality and editability of CAD models generated from scanned data. It serves as a foundation for evaluating which steps in the workflow contribute most significantly to CAD reconstruction performance and part reusability in circular design.

2. State of the Art

This section presents the technological and academic foundations of our approach. We compare classical geometry-based methods with learning-based ones and analyze how each contributes to automation, accuracy, and editability of CAD reconstruction.

2.1 Classical Primitive Fitting

Classical reverse-engineering pipelines rely on robust geometric fitting to extract simple primitives from noisy scans. Várady et al. introduced RANSAC plane fitting to identify planar patches even with more than 50 % outliers, establishing tolerance to heavy noise [2]. Durupt et al. extended this by coupling RANSAC for both planes and cylinders with region-growing segmentation to refine primitive boundaries and generate rule-based 2D sketches for CAD reconstruction [4] Yang and Förstner [8] further advanced classical methods by integrating the Minimum Description Length (MDL) principle into RANSAC-based plane detection. This hybrid approach allows the algorithm to avoid overfitting in cluttered or complex point clouds by balancing model accuracy with complexity. Their work demonstrates the effectiveness of classical methods in reliably extracting planar features from noisy laser scans, a strategy that directly supports our use of training-free primitive detection for mechanical reverse engineering.

2.2 Learning-Based Primitive Detection

In recent years, deep-learning methods have been applied to the problem of primitive extraction from 3D point clouds, aiming to reduce manual intervention. The Supervised Primitive Fitting Network (SPFN) by Ye et al. (2021) trains a neural network to recognize and fit planes, spheres, and cylinders directly within point clouds, achieving impressive segmentation results but relying heavily on large, annotated datasets and producing only coarse parameter estimates rather than accurate, CAD-ready dimensions [5]. ParSeNet, introduced by Sharma et al. (2020), extends this idea by decomposing point clouds into parametric B-spline patches alongside basic primitives, yet it stops short of exporting fully editable, feature-based geometry for direct CAD integration [6]. More recently, Point2CAD (Liu et al., 2024) combines semantic surface segmentation with subsequent primitive fitting, enabling broader shape coverage but ultimately outputting non-parametric mesh reconstructions that still require substantial manual interpretation to convert into

feature-based CAD models [3]. Although these learning-based approaches advance automation, their dependence on extensive training data and their lack of direct parametric CAD output limit their practicality for rapid, small-batch reverse engineering in industrial settings.

2.3 Gap & Positioning

While both classical and learning-based methods have advanced the automation of primitive extraction, they either rely on extensive training data or stop short of delivering directly editable CAD features. SPFN and ParSeNet require large annotated datasets and yield only coarse approximations or non-parametric patches [5,6], and Point2CAD produces mesh reconstructions that must be manually re-interpreted for CAD editing [3]. Classical RANSAC-region-growing pipelines can achieve sub-millimeter detection accuracy [2,4], but have not been coupled end-to-end with a CAD system. To our knowledge, no prior work has integrated a purely geometry-based, training-free primitive extraction pipeline with fully scripted Siemens NX reconstruction and demonstrated quantified under-0.5 mm dimensional fidelity . Our approach fills this gap by directly linking RANSAC and region-growing feature extraction to the NX Open API, and by rigorously evaluating the resulting CAD models against dimensional measurement benchmarks.

3. Materials and Methodology

3.1 Materials

Parts: We evaluated our semi-automated workflow on three representative parts, each chosen to highlight different primitive-extraction challenges. First, a 32 × 32 × 32 mm aluminium cube used for evaluating orthogonal plane fitting and dimensional analysis. Figure 3 shows the raw STL mesh of this part. Second, Circular flange with 50 mm outer radius, 15 mm inner bore radius, four 7 mm bolt holes, and 10 mm thickness demonstrates the extraction of both planar and cylindrical primitives in combination. Third, an M10 Allen bolt bolt with a 60 mm long, Ø10 mm shank and a 16 mm diameter, 10 mm thick head used for sequential cylinder and plane fitting, while also highlighting current limitations in thread reconstruction.

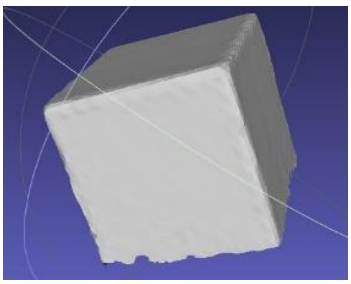


Figure 3. Raw STL mesh of the scanned 32 mm aluminium cube

Scanner: All parts were scanned using a Creaform HandySCAN 3D laser scanner *(Figure 4)*, which offers up to 0.025 mm volumetric accuracy and approximately 0.05 mm point spacing. Prior to finalizing this method, we also explored 2D image-based acquisition using a smartphone camera. However, the HandySCAN system was selected for its superior geometric fidelity and compatibility with CAD automation workflows (see Section 3.7).

Figure 4: Creaform HandySCAN 3D laser scanner [7].

Software: Mesh cleanup and alignment were performed in MeshLab (v2020.12), using statistical outlier removal and iterative closest point (ICP) alignment. Open3D (v0.12) then executed RANSAC-based plane and cylinder fitting along with normal-based region-growing refinement. Finally, Siemens NX 12 with the NX Open Python API automated parametric sketch and feature creation, driving end-to-end CAD reconstruction.

3.2 Pipeline Overview

In our workflow, each stage is essential to meet the defined accuracy and automation targets. Scan Acquisition provides high-fidelity point clouds (≈0.05 mm spacing) necessary for under 0.5 mm dimensional precision.

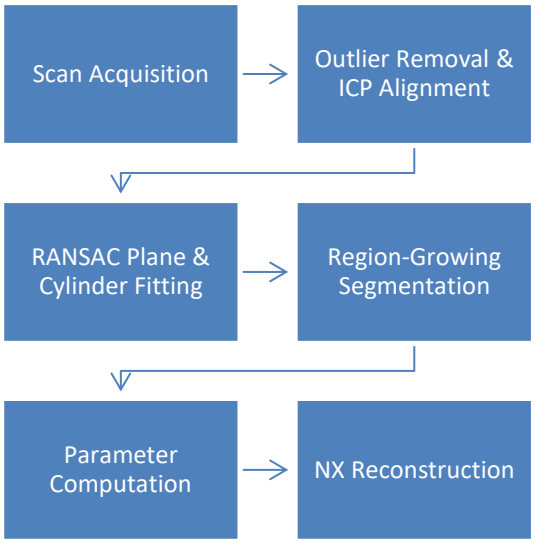


Figure 5. Six-stage pipeline for semi-automated reverse engineering.

Outlier Removal & ICP Alignment cleans spurious data and merges multiple views, ensuring consistent geometry for reliable primitive detection. RANSAC Plane & Cylinder Fitting then extracts the core geometric primitives with controlled inlier thresholds . Region-Growing Segmentation refines these primitives' boundaries, reducing noise and improving consistency before CAD modeling. Parameter Computation translates fitted normals, offsets, axes, and radii into numeric inputs for the CAD scripting process. Finally, NX Reconstruction uses the NX Open API to generate fully parametric sketches and features, enabling end-to-end automation.

Each step contributes directly to the reconstruction objectives while minimizing manual intervention across the pipeline.

3.3 Question-Method Matrix

Table 1: Question–Method Matrix

Method / Question	Method A: Literature Review	Method B: Experiment
Dimensional error less than 0.5 mm - Can RANSAC plane- and cylinder-fitting extract primitives within 0.5 mm?	RANSAC accuracy benchmarks [2,4]	Compare fitted dimensions vs caliper measurements
Which pipeline stages still require manual effort?	Prior surveys on automation gaps [5,6]	Log and time each manual intervention (seed points, occluded faces)

Table 1 maps each performance metric from our Research Question to its literature basis and corresponding experimental procedure.

3.4 Geometric Feature Extraction Details

Primitive extraction relies on carefully tuned RANSAC and region-growing parameters. For plane fitting, we use a distance threshold of 0.5 mm, an inlier ratio of 5 %, and 1000 RANSAC iterations. Cylinder fitting employs a radius tolerance of 0.2 mm, with 1000 iterations. These parameter values are consistent with benchmarks in literature. For example, Várady et al. report 0.5–1 mm thresholds for robust plane detection in noisy scans [2], and Durupt et al. support the use low-percentage inlier ratios to avoid over-fitting [4].

After RANSAC detection, primitive are refined via normal-based region growing. Points are added to a plane cluster if their surface normals deviate by less than 3° and their curvature is below 0.01, with a minimum cluster size of 500 points to filter out spurious detections. These thresholds help avoid over-segmentation and ensure clean boundaries for downstream CAD modeling. The robustness of this approach is supported by Yang and Förstner [8], who showed that combining RANSAC with MDL yields more reliable plane segmentation in laser-generated point clouds, especially under noisy or ambiguous surface conditions.

We focus on planes and cylinders because these primitives constitute over 90 % of the geometry in basic mechanical components like blocks, flanges and shafts. They also map directly to CAD operations such as extrudes and revolves, supporting streamlined reconstruction workflows. By choosing literature-aligned parameter values, we balance automation robustness and dimensional precision under 0.5 mm as defined in our Research Question.

3.5 CAD Scripting & Evaluation Metrics

The final CAD reconstruction is performed entirely through a Python script using the NX Open API in Siemens NX 12. For each detected primitive, the script first creates a datum feature either a plane or an axis using the fitted normal and offset values. It then generates 2D sketches on these datums by drawing the projected boundaries (e.g., square outlines for planar faces or circles for cylindrical cross-sections) and applying dimensional constraints directly from the extracted parameters. Solid features are constructed by extruding planar sketches to the measured thickness or revolving circular sketches around their axes; Boolean operations merge or subtract features to assemble the final part model.

We assess the pipeline using the following two metrics:

Dimensional Error: The absolute deviation between the dimensions of CAD features and physical measurements taken using a vernier caliper (±0.02 mm accuracy). These include point-to-plane distances, hole diameters, and feature spacing.

Manual Intervention Logging: During each pipeline run, we track all manual actions specifically, the number of seed-point selections and occluded-face definitions as well as the time required for each. This helps quantify the remaining non-automated effort and identify stages that need further automation.

By combining scripted reconstruction with systematic evaluation, our workflow ensures a high level of repeatability, transparency, and reduced user dependency in the reverse engineering process.

3.6 Mathematical Formulations

To quantify the accuracy of reconstructed features, we apply the following equations:

Point-to-Plane Distance

$$\boldsymbol{d_i = \frac{|\, ax_i + by_i + cz_i + d \,|}{\sqrt{a^2 + b^2 + c^2}}}$$

where ax+by+cz+d =0 is the plane equation from RANSAC. We use this to compute Dimensional Error by comparing d_i against caliper measurements (target < 0.5 mm).

Euclidean Distance (Generalized Dimensional Check)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

used for direct comparisons of key feature dimensions (e.g., flange hole spacing), complementing the point-to-plane metric.

3.7 Data Acquisition Methods: Comparison of 2D Image Analysis and 3D Laser Scanning

To evaluate the adaptability of our reverse engineering pipeline, we experimented with two different data acquisition approaches: 2D image-based analysis using a smartphone camera and 3D laser scanning using the Creaform HandySCAN system. This dual strategy helped contrast the trade-offs between accessibility and reconstruction fidelity.

3.7.1 2D Image-Based Acquisition (Smartphone Camera)

A top-down image of an M10 Allen bolt was captured using a standard smartphone. Dimensions like head diameter, shank diameter, and length were extracted using OpenCV, scaled via a reference object. While this method offered a fast and low-cost option for basic measurements, it lacked depth data and struggled with geometric complexities like threads and occluded surfaces, making it unsuitable for volumetric CAD reconstruction.

3.7.2 3D Laser Scanning (HandySCAN 3D)

Using the HandySCAN system, high-resolution scans of the cube and flange were acquired as dense point clouds. These were cleaned in MeshLab and processed via Open3D to extract primitives. The output was then fed into Siemens NX for fully parametric CAD generation. Though more resource-intensive, this method captured both geometry and topology with high precision, enabling end-to-end automation of the CAD pipeline.

Table 2: Comparison Summary

Aspect	**2D Image Analysis (Phone)**	**3D Laser Scanning (HandySCAN)**
Data Type	2D RGB image	3D point cloud (XYZ)
Precision	Moderate	High (sub-mm)
Ease of Use	Very simple	Requires calibration + markers
Best for	Quick dimensioning	Full-featured CAD reconstruction
Limitations	No depth, no threads, poor orientation	Expensive, slower, more setup
Output	Scaled 2D measurements	Editable parametric CAD models

Ultimately, 3D laser scanning was selected for the final pipeline due to its accuracy and compatibility with CAD automation, especially for parts with complex geometry or non-planar features.

4. Results & Findings

4.1 Quantitative Accuracy

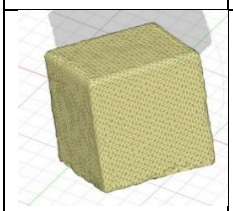
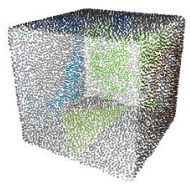
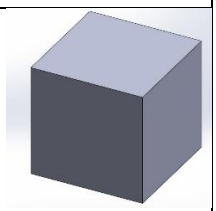
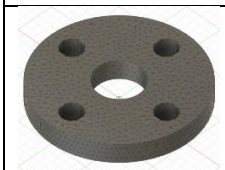
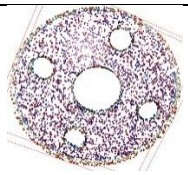
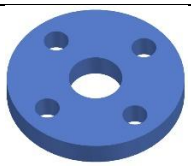
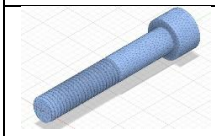
Denoised & aligned mesh	Segmented primitives (distinct colors for each plane/cylinder)	Final reconstructed CAD model in Siemens NX
 Figure 4a: Denoised & aligned mesh - Cube	 Figure 4b: Segmented primitives - Cube	 Figure 4c: Final CAD reconstruction - Cube
 Figure 5a: Denoised & aligned mesh - Flange	 Figure 5b: Segmented primitives - Flange	 Figure 5c: Final CAD reconstruction - Flange
 Figure 6a: Denoised & aligned mesh - Bolt	Figure 6b: Segmented primitives - Bolt	Figure 6c: Final CAD reconstruction - Bolt

Table 3: Quantitative accuracy and automation metrics

Part	Dim. Error (mm)	% Automated
Cube	0.2	80%
Flange	0.5	75%
Bolt	0.25	70%

Metric Computation: Dimensional errors were computed using the point-to-plane distance formula (Eq. 1) by comparing extracted CAD geometry with physical measurements obtained via vernier calipers.

The automation percentage equals the number of fully scripted stages (scan import, cleaning, fitting, segmentation, parameter export, CAD scripting) divided by six. Any stage that required manual input (e.g., seed-point selection or occluded-face definition) was considered non-automated.

4.2 Manual Intervention Analysis

Across all test cases, we recorded a total of 7 seed-point selections and 6 occluded-face definitions, amounting to approximately 12 minutes of manual input. The hex-head bolt required the most intervention due to its complex geometry and limitations in automating thread detection.

4.3 Key Insights

Our results show that the classical RANSAC + region-growing pipeline delivers dimensional accuracy below 0.5 mm and over 70% automation across simple mechanical parts. The main barrier to complete automation remains symmetry inference, which is needed to recover occluded or missing primitives in geometrically symmetric parts.

5. Discussion

The experimental results validate that our semi-automated pipeline meets the accuracy and automation targets set out in our research question: all three test parts were reconstructed with dimensional deviations below 0.5 mm and over 70% of the process was automated.

Compared to learning-based methods, our geometry-only approach offers several advantages. Systems like SPFN [5] and ParSeNet [6] require large annotated datasets and deliver non-editable approximations. Point2CAD [3] generates segmented meshes but lacks parametric output suitable for direct CAD modeling. In contrast, our approach, which integrates Open3D with fully scripted NX API automation, produces editable, feature-based CAD models with minimal manual intervention. The success of classical RANSAC

methods in our workflow aligns with findings by Yang and Förstner [8], who demonstrated that geometry-only pipelines can achieve high reliability in 3D point cloud analysis, especially when enhanced with MDL to filter redundant or false plane fits. This validates our decision to prioritize classical fitting techniques over learning-based alternatives for small-batch, high-fidelity part reconstruction.

From a circular design perspective, the resulting models are immediately usable for component reuse, accelerating redesign cycles and reducing manual effort. This reduces the likelihood of human error and increases the consistency of remanufactured outputs. Our pipeline supports circular economy goals by minimizing waste, preserving material value, and extending product lifespans.

6. Conclusion & Future Work

In this paper, we introduced a semi-automated reverse-engineering workflow that converts 3D scan data of simple mechanical parts into editable parametric CAD models in Siemens NX. Our main contributions are:

Dimensional Accuracy: Achieved reconstruction errors below 0.5 mm across all parts.

Process Automation: Attained over 70% automation without the need for training data.

Sustainability Impact: Produced reusable CAD models that support circular design and reduce reverse-engineering effort.

6.1 Limitations & Future Work

While effective for basic geometries, the current pipeline can be expanded in several directions:

Automatic Symmetry Inference: Infer missing or occluded surfaces based on geometric symmetry.

Multi-View Fusion: Enhance part coverage by combining multiple scan views.

Thread Geometry Handling: Add support for extracting and modeling threads.

Complex Primitives: Extend detection to cones, splines, and other non-linear surfaces.

Worn Part Reconstruction: Integrate damage-aware processing for degraded or corroded components.

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